# Fusion, Reranking and Metrics

## 1 Combining retrievers

Reciprocal rank fusion is a method that combines several ranked lists by summing one over a constant plus the rank of each passage. Hybrid retrieval combines dense retrieval and BM25 to cover both meaning and exact terms. A cross-encoder is a model that reads the query and a passage together and scores their relevance. Reranking reorders the top passages with a slower and more accurate model.

## 2 Measuring retrieval

Recall at k is the share of the relevant passages that appear in the top k results. Precision at k is the share of the top k results that are relevant. F1@k: The harmonic mean of precision and recall RankNet is a pairwise loss that measures the correctness of relative passage orders Reranking can improve precision at k but it cannot recover a passage the first stage never retrieved.